

RFM Customer Retention Strategy

1. Project Overview

A UK-based online retailer sells household and office products. Between Dec 2010 and Dec 2011, management observed declining repeat purchase rates and wanted to shift from a one-size-fits-all marketing approach to segmented, data-driven retention.

Business questions:

1. Which customers segments drive the most revenue?
2. Which segments carry the highest churn risk (90+ days without a purchase)?

Using RFM (Recency, Frequency, Monetary) quintile scoring, 4,338 customers were segmented into seven behavioural groups. The output is a tiered retention strategy that directs marketing resources to the highest-impact segments first.

2. Dataset Summary

- Source: <https://archive.ics.uci.edu/dataset/352/online+retail>
- Rows: 541,909
- Columns: 8
- Period: Dec 2010 - Dec 2011
- Key Features:
 - Transaction details (InvoiceNo, StockCode, Quantity, UnitPrice)
 - Customer info (CustomerID, Country)
 - Product info (Description)
 - Temporal (InvoiceDate)
 - Missing Data: 135,080 values in CustomerID column

3. Exploratory Data Analysis using Python

Data preparation and cleaning were performed in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```

<class 'pandas.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   invoice_no      541909 non-null  str    
1   stock_code      541909 non-null  str    
2   product_name    540455 non-null  str    
3   quantity        541909 non-null  int64  
4   invoice_date    541909 non-null  str    
5   unit_price      541909 non-null  float64
6   customer_id     406829 non-null  float64
7   country         541909 non-null  str    
dtypes: float64(2), int64(1), str(5)
memory usage: 67.8 MB

```

	quantity	unit_price	customer_id
count	541909.000000	541909.000000	406829.000000
mean	9.552250	4.611114	15287.690570
std	218.081158	96.759853	1713.600303
min	-80995.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13953.000000
50%	3.000000	2.080000	15152.000000
75%	10.000000	4.130000	16791.000000
max	80995.000000	38970.000000	18287.000000

4. Data Cleaning

- **Date Parsing:** Converted invoice_date to datetime with dayfirst=True
- **Removed Cancellations:** Filtered out all invoices where invoice_no starts with 'C'
- **Removed Missing Customers:** Dropped rows where customer_id is null
- **Removed Invalid Prices:** Filtered rows where unit_price <= 0
- **Removed Invalid Quantities:** Filtered rows where quantity <= 0
- **Created Revenue Column:** revenue = quantity * unit_price
- **Reference Date:** Set to 2011-12-09 (latest transaction in the dataset)

Final cleaned shape: 397884 rows * 9 columns

5. RFM Scoring & Segmentation

- **Aggregation:** Grouped by customer_id to compute Recency, Frequency and Monetary per customer.
- **Scoring:** Applied pd.qcut() with 5 bins (quintiles) after rank(method='first') to handle ties.
- **Segmentation:** Assigned 7 named segments via a rule-based decision tree on RFM scores.
- **Churn Flag:** churned = 1 if recency $\geq$ 90 days.

Churn threshold justification: A 90-day inactivity threshold was chosen based on the dataset's median inter-purchase interval (~45 days) - customers inactive for 2x the median gap are considered churned.

Segmentation hierarchy:

- 1) Recency $\rightarrow$ splits ACTIVE ($R \geq 3$) vs LAPSED ($R \leq 2$)
- 2) Frequency $\rightarrow$ ranks loyalty within each tier
- 3) Monetary $\rightarrow$ separates high-value from low-value

Segment	Customers	Criteria	Business Meaning
Champions	947	$R \geq 4, F \geq 4, M \geq 4$	Recent, frequent, high-spend. Protect all costs.
Loyal Customers	454	$R \geq 3, F \geq 4, M \geq 3$	Recent, very frequent, solid spend. Reliable revenue.
Potential Loyalists	190	$R \geq 4, F = 3, M \geq 3$	Recent, moderate frequency, good spend. Nudge toward loyalty.
Can't Lose Them	173	$R \leq 2, F \geq 4, M \geq 4$	Lapsed, but was high-value & frequent. Highest urgency.
At Risk	207	$R = 2, F \geq 3, M \geq 3$	Recently lapsed, was a decent buyer. Recoverable window.
Hibernating	106	$R \leq 2, M \geq 4$	Lapsed, low frequency, but high per-transaction value. Low-cost nudge.
Low Priority	2,261	All remaining	Insufficient value or engagement for active intervention.

6. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in PostgreSQL to answer key business questions:

1) **Revenue by Segment** – Identified the revenue contribution of each customer segment.

	segment text	total_customers bigint	total_revenue numeric	revenue_share_pct numeric	avg_revenue_per_customer numeric
1	Champions	947	5755513.22	64.6	6077.63
2	Low Priority	2261	926397.37	10.4	409.73
3	Loyal Customers	454	908853.14	10.2	2001.88
4	Can't Lose Them	173	390109.94	4.4	2254.97
5	Potential Loyalists	190	370934.81	4.2	1952.29
6	Hibernating	106	346143.89	3.9	3265.51
7	At Risk	207	213455.53	2.4	1031.19

2) **Customer Distribution**– Showed the proportion of customers in each segment.

	segment text	customers bigint	pct_of_total numeric
1	Low Priority	2261	52.1
2	Champions	947	21.8
3	Loyal Customers	454	10.5
4	At Risk	207	4.8
5	Potential Loyalists	190	4.4
6	Can't Lose Them	173	4.0
7	Hibernating	106	2.4

3) **Churn Rate & Revenue at Risk** – Measured the percentage of churned customers and the monetary value at risk per segment.

	segment text	total_customers bigint	churned_customers numeric	churn_rate_pct numeric	total_revenue numeric	revenue_at_risk numeric
1	Low Priority	2261	1107	49.0	926397.37	347716.01
2	Hibernating	106	85	80.2	346143.89	305218.21
3	Can't Lose Them	173	114	65.9	390109.94	244943.31
4	At Risk	207	143	69.1	213455.53	137392.88
5	Champions	947	0	0.0	5755513.22	0.00
6	Potential Loyalists	190	0	0.0	370934.81	0.00
7	Loyal Customers	454	0	0.0	908853.14	0.00

- 4) **Top 10 High-Value At-Risk Customers** – Identified the 10 churned customers with the highest historical spend (90th percentile threshold) to prioritise for win-back outreach.

	customer_id bigint	segment text	recency bigint	frequency bigint	monetary numeric
1	12346	Hibernating	325	1	77183.60
2	15749	Hibernating	234	3	44534.30
3	15098	Hibernating	181	3	39916.50
4	16180	Can't Lose Them	99	8	10254.18
5	12590	Hibernating	210	2	9864.26
6	13093	Can't Lose Them	275	8	7832.47
7	12980	Can't Lose Them	157	9	7374.90
8	13027	Can't Lose Them	113	6	6912.00
9	16553	Can't Lose Them	162	12	5719.82
10	17850	Can't Lose Them	371	34	5391.21

- 5) **Top 3 Products by Revenue per Segment** – Identified cross-sell opportunities for Loyal Customers and Champions.

	segment text	product_name text	product_revenue numeric	rank bigint
1	At Risk	WHITE HANGING HEART T-LIGHT HOLD...	3459.05	1
2	At Risk	EMPIRE DESIGN ROSETTE	3215.42	2
3	At Risk	REGENCY CAKESTAND 3 TIER	2928.75	3
4	Can't Lose Them	PARTY BUNTING	8508.05	1
5	Can't Lose Them	REGENCY CAKESTAND 3 TIER	7571.20	2
6	Can't Lose Them	SET/4 WHITE RETRO STORAGE CUBES	5042.80	3
7	Champions	REGENCY CAKESTAND 3 TIER	104159.85	1
8	Champions	WHITE HANGING HEART T-LIGHT HOLD...	66637.30	2
9	Champions	JUMBO BAG RED RETROSPOT	60945.25	3
10	Hibernating	MEDIUM CERAMIC TOP STORAGE JAR	77283.60	1
11	Hibernating	PICNIC BASKET WICKER 60 PIECES	39619.50	2
12	Hibernating	FAIRY CAKE FLANNEL ASSORTED COLO...	13193.55	3
13	Low Priority	POSTAGE	12870.90	1
14	Low Priority	REGENCY CAKESTAND 3 TIER	12480.15	2
15	Low Priority	Manual	10409.92	3
16	Loyal Customers	Manual	15590.08	1
17	Loyal Customers	POSTAGE	14002.70	2
18	Loyal Customers	REGENCY CAKESTAND 3 TIER	12094.50	3
19	Potential Loyalists	PAPER CRAFT , LITTLE BIRDIE	168469.60	1
20	Potential Loyalists	RABBIT NIGHT LIGHT	7208.90	2
21	Potential Loyalists	JUMBO BAG RED RETROSPOT	2305.57	3

- 6) **Average Purchase Gap per Segment** – Measured the average number of days between consecutive purchases.

	segment text	avg_days_between_purchases numeric
1	Hibernating	0.8
2	Champions	1.1
3	Low Priority	1.6
4	Can't Lose Them	1.7
5	Loyal Customers	2.1
6	Potential Loyalists	2.1
7	At Risk	2.7

- 7) **Monthly Revenue Trend (Dec 2010 – Dec 2011)** – Confirmed a strong Q4 seasonal spike, suggesting some customers classified as churned may be seasonal rather than permanently lost.

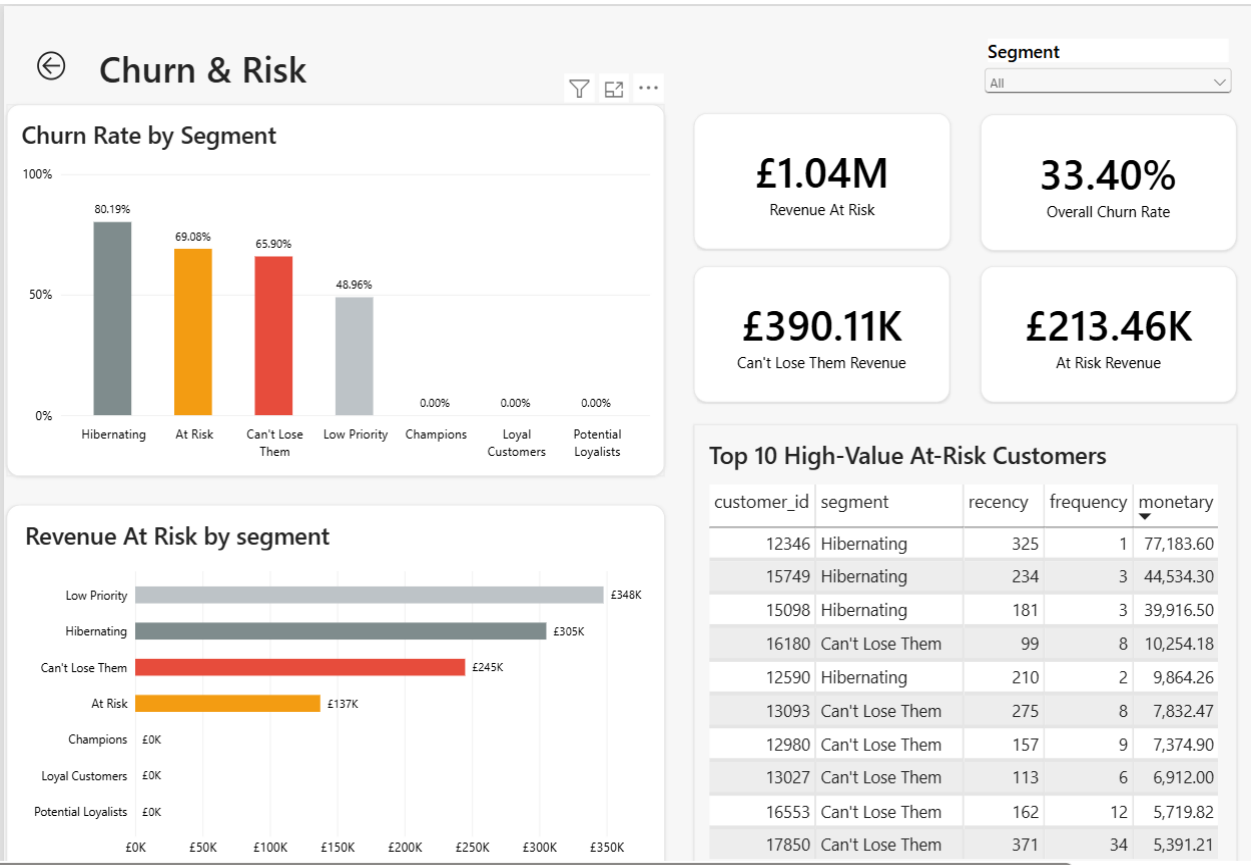
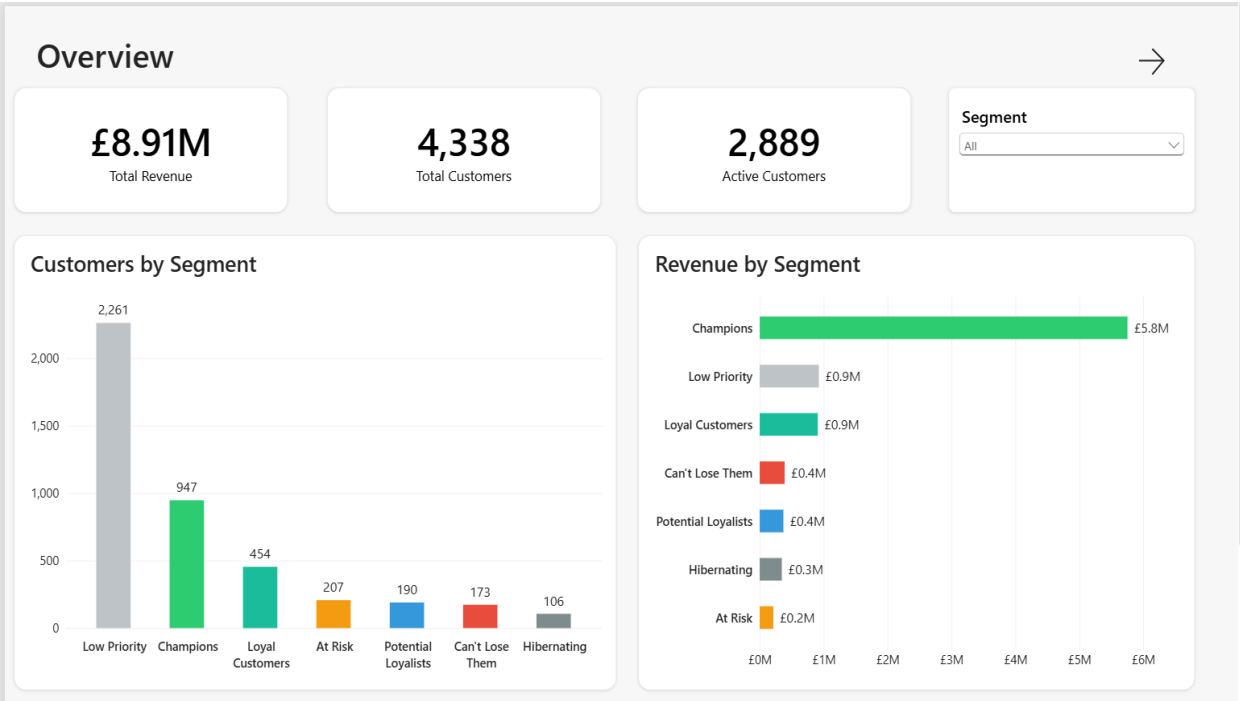
	month timestamp without time zone	active_customers bigint	monthly_revenue numeric
1	2010-12-01 00:00:00	885	572713.89
2	2011-01-01 00:00:00	741	569445.04
3	2011-02-01 00:00:00	758	447137.35
4	2011-03-01 00:00:00	974	595500.76
5	2011-04-01 00:00:00	856	469200.36
6	2011-05-01 00:00:00	1056	678594.56
7	2011-06-01 00:00:00	991	661213.69
8	2011-07-01 00:00:00	949	600091.01
9	2011-08-01 00:00:00	935	645343.90
10	2011-09-01 00:00:00	1266	952838.38
11	2011-10-01 00:00:00	1364	1039318.79
12	2011-11-01 00:00:00	1664	1161817.38
13	2011-12-01 00:00:00	615	518192.79

7. Dashboard in Power BI

An interactive two-page dashboard was built in Power BI:

- Page 1 (Overview): Total revenue, customer count, active customers, revenue by segment and customer by segment.

- Page 2 (Churn & Risk): Revenue at risk, churn rate, segment-level risk metrics and a prioritised list of top 10 high-value churned customers.



8. Key Findings

- Champions + Loyal Customers (1,401 customers, 32.3%) generate 74.9% (£6.67M) of total revenue.
- £1,035,269 in revenue is at risk across all segments (customers inactive 90+ days).
- **Can't Lose Them** is the highest-urgency win-back target. 173 customers, 93% churn rate, £398K already lost.
- **At Risk** is the most preventable churn. 207 customers, 68% churn rate, £241K at risk ,still within a recoverable window.
- 52.1% of customers are Low Priority (2,261 out of 4,338). Marketing spend should focus on the remaining 2,077 customers.
- Oct–Nov shows a seasonal spike (Black Friday / pre-Christmas). Some churned customers may be seasonal buyers.

9. Recommendations

	Segment	Action	Priority
1	Champions	Implement a loyalty tier program or offer early access to new products/sales to reward top customers	P2 – Protect
2	Loyal Customers	Cross-sell complementary product categories based on past purchase behaviour to increase basket size	P2 – Protect
3	Potential Loyalists	Create bundle offers or subscription incentives to increase purchase frequency and solidify their loyalty	P3 – Nurture
4	Can't Lose Them	Launch a personalised win-back campaign with exclusive high-value offers or discounts to prevent churn	P1 – Act now
5	At Risk	Automate a “we miss you” nudge at day 60 of inactivity, offering a small incentive to re-engage	P1 – Act now
6	Hibernating	Send low-cost email campaigns with personalized high-value product recommendations or content relevant to their past interests	P3 – Nurture
7	Low-Priority	Supress these customers from high-cost marketing campaigns to optimize spend and focus resources on higher-value segments	P4 – Deprioritize

Priority logic:

P1 – Act now: High revenue at risk — inaction means losing the most money.

P2 – Protect: High revenue, stable — reward to maintain.

P3 – Nurture: Moderate potential, low urgency — build over time.

P4 – Deprioritize: Not worth the spend — free resources for higher-value segments.

10. Limitations

- 90-day churn threshold is a business assumptions, not data-driven.
- No external factors (marketing spend, economic conditions) considered.
- Segment boundaries are rule-based, adjacent quintiles may produce similar profiles.
- Data covers only 12-month period; Oct–Nov spike may misclassify seasonal buyers as churned.